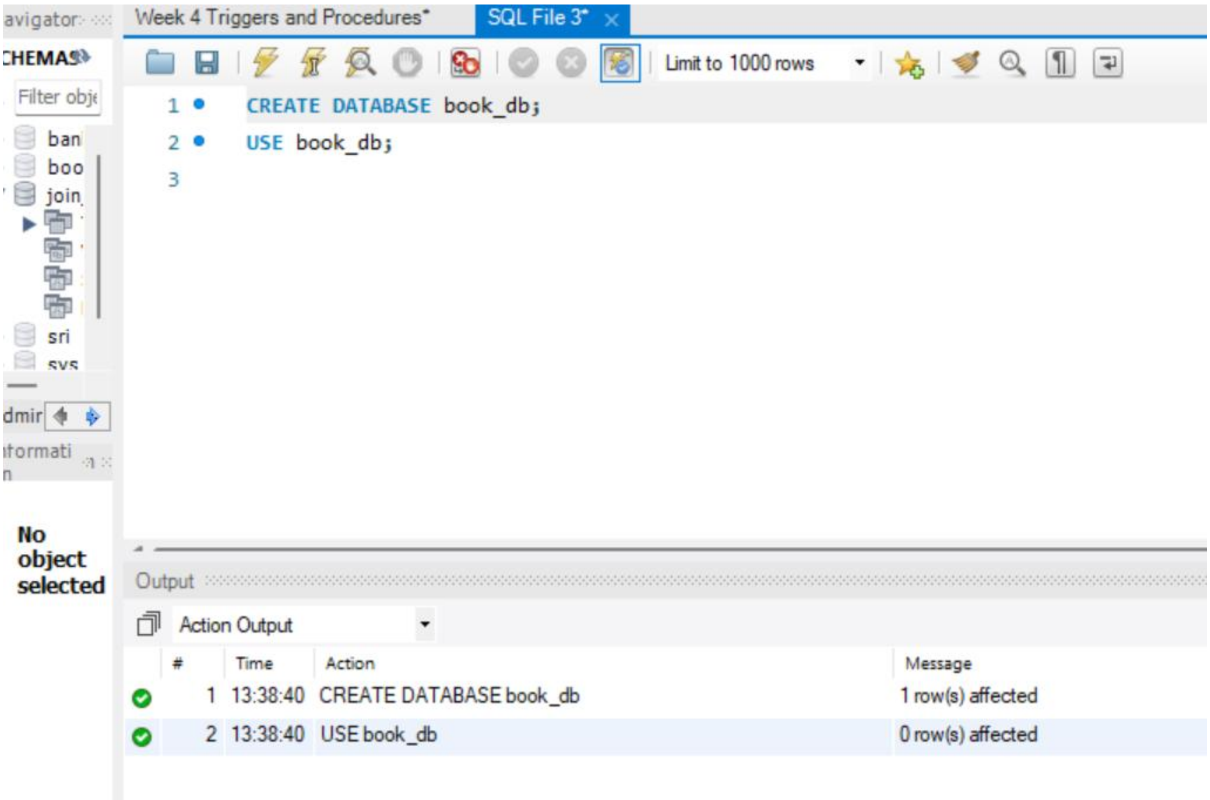
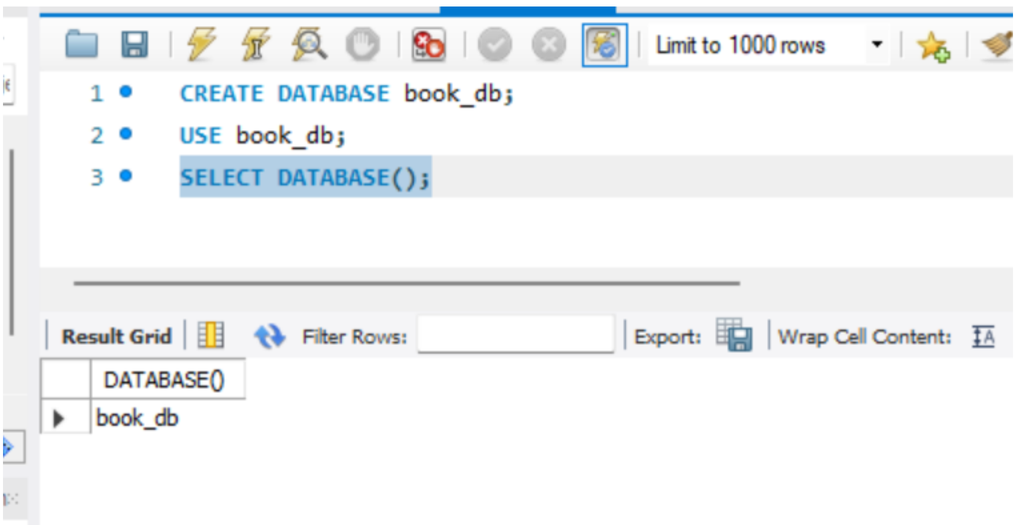
Practical 4

CREATE DATABASE bookflow_db;

USE bookflow_db;

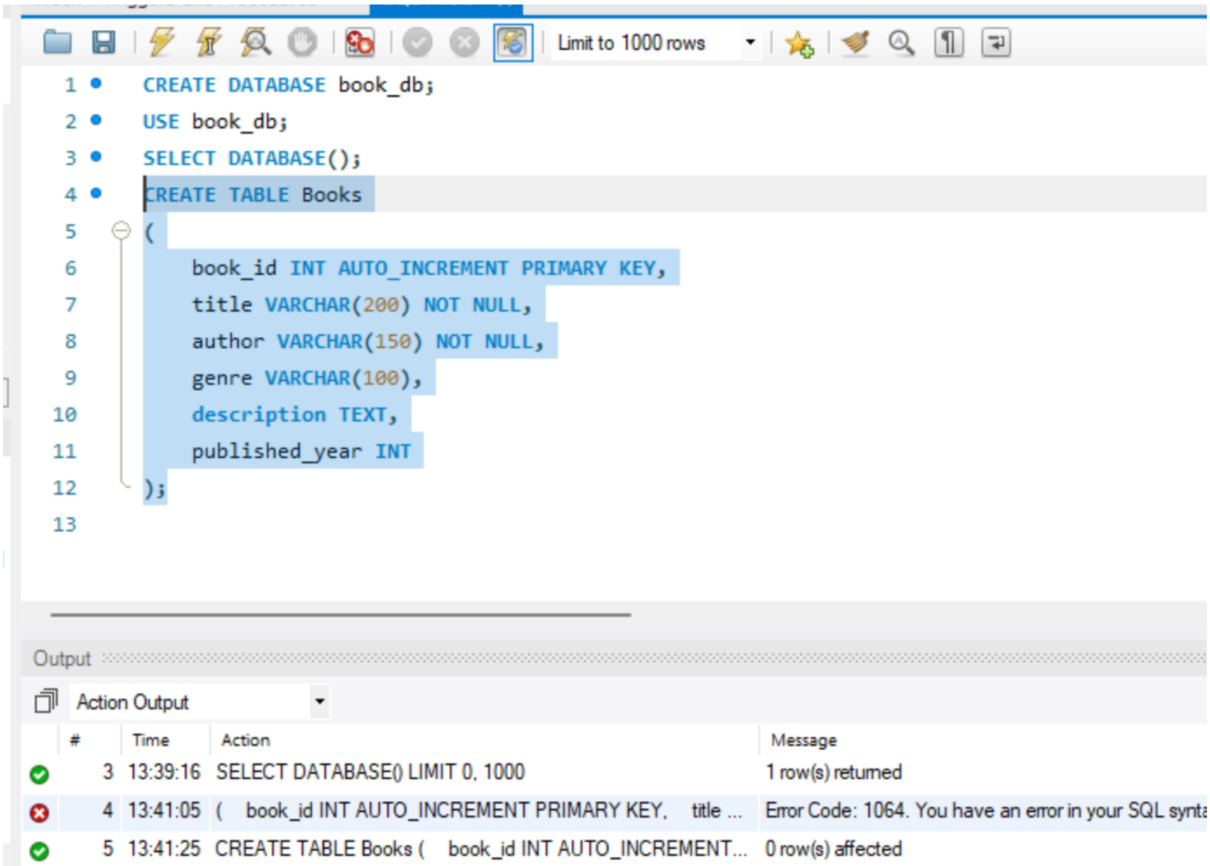


SELECT DATABASE();

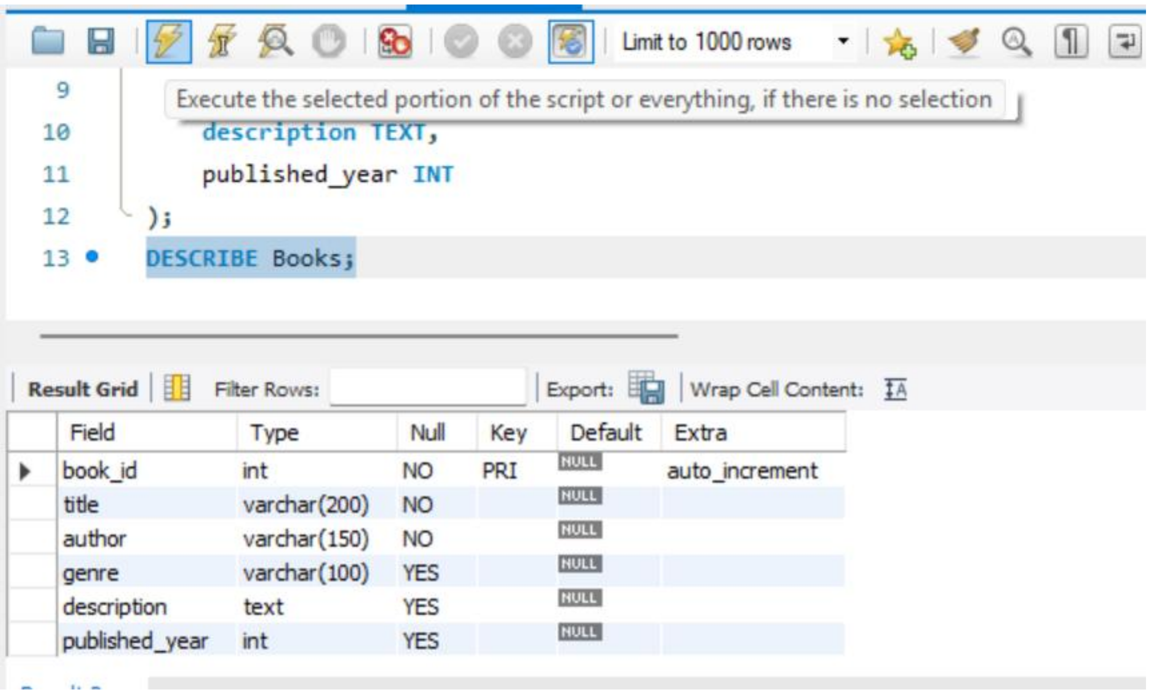


CREATE TABLE Books

```
(
    book_id INT AUTO_INCREMENT PRIMARY KEY,

    title VARCHAR(200) NOT NULL,

    author VARCHAR(150) NOT NULL,

    genre VARCHAR(100),

    description TEXT,

    published_year INT
);
```



DESCRIBE Books;

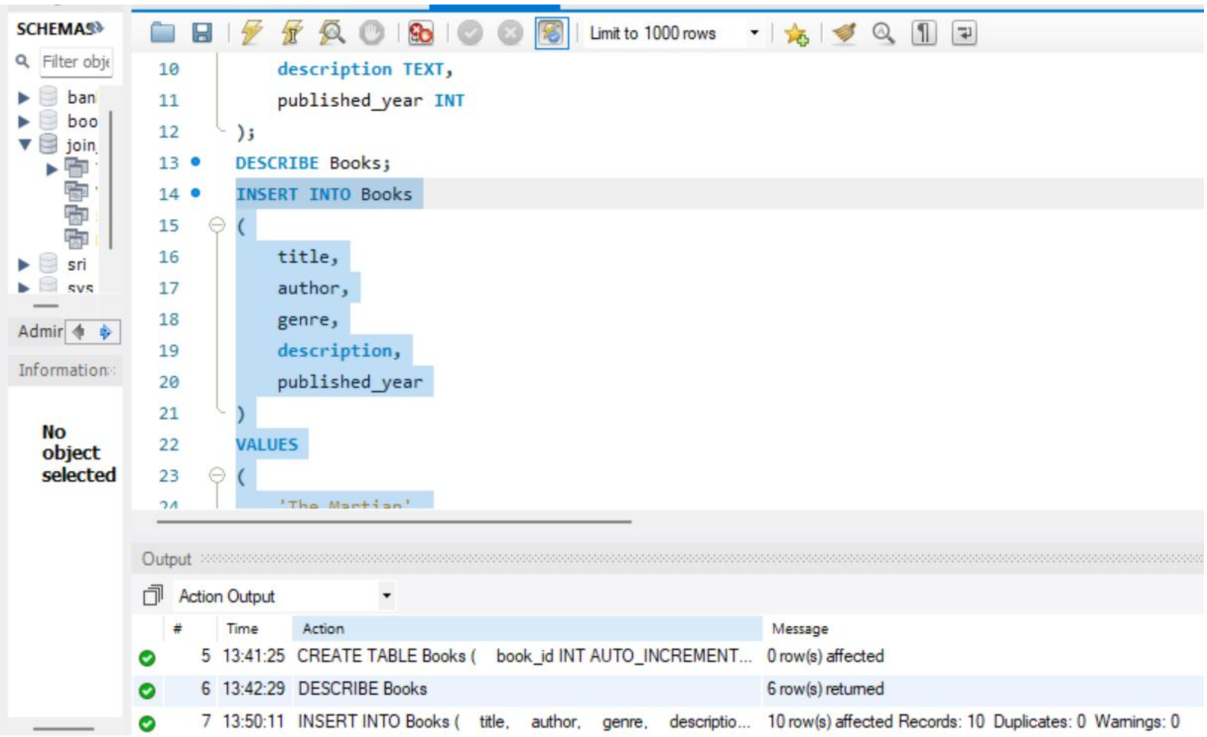



```
INSERT INTO Books
(
  title,
  author,
  genre,
  description,
  published_year
)
VALUES

(
  'The Martian',
  'Andy Weir',
  'Science Fiction',
  'An astronaut struggles to survive alone on Mars.',
  2011
),
```

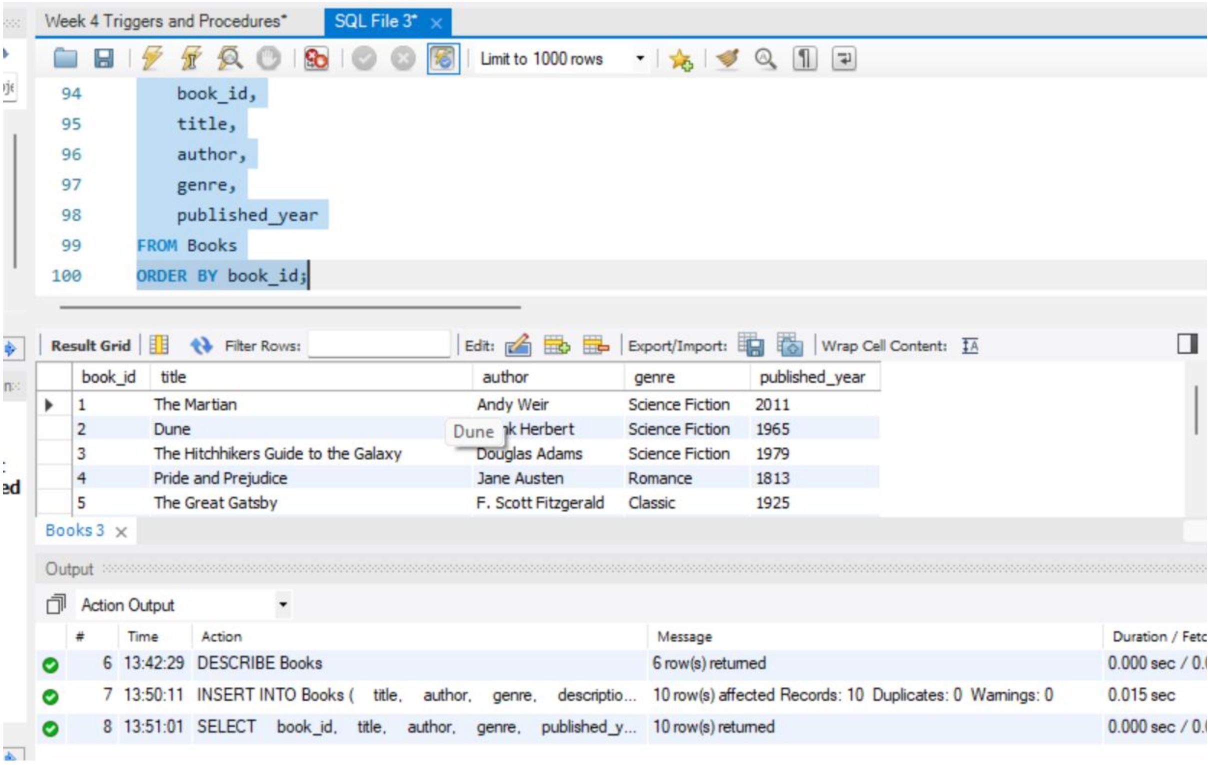
```
(
    'Dune',
    'Frank Herbert',
    'Science Fiction',
    'A story involving planets, politics, survival and space.',
    1965
),

(
    'The Hitchhikers Guide to the Galaxy',
    'Douglas Adams',
    'Science Fiction',
    'A humorous adventure across space and alien worlds.',
    1979
),

(
    'Pride and Prejudice',
    'Jane Austen',
    'Romance',
    'A classic story about relationships and social expectations.',
    1813
),

(
    'The Great Gatsby',
    'F. Scott Fitzgerald',
    'Classic',
    'A story of ambition, wealth, love and the American dream.',
    1925
),
```

```
(
   'Foundation',
   'Isaac Asimov',
   'Science Fiction',
   'A futuristic story about mathematics, civilization and space.',
   1951
),

(
   'The Hobbit',
   'J.R.R. Tolkien',
   'Fantasy',
   'A fantasy adventure involving Bilbo Baggins and a dangerous journey.',
   1937
),

(
   'Enders Game',
   'Orson Scott Card',
   'Science Fiction',
   'A young student is trained for an interstellar conflict.',
   1985
),

(
   'Harry Potter and the Sorcerers Stone',
   'J.K. Rowling',
   'Fantasy',
   'A young wizard begins his magical education.',
   1997
```

```
),

(
  'Twenty Thousand Leagues Under the Seas',
  'Jules Verne',
  'Adventure',
  'A fantastic underwater adventure aboard a mysterious submarine.',
  1870
);
```

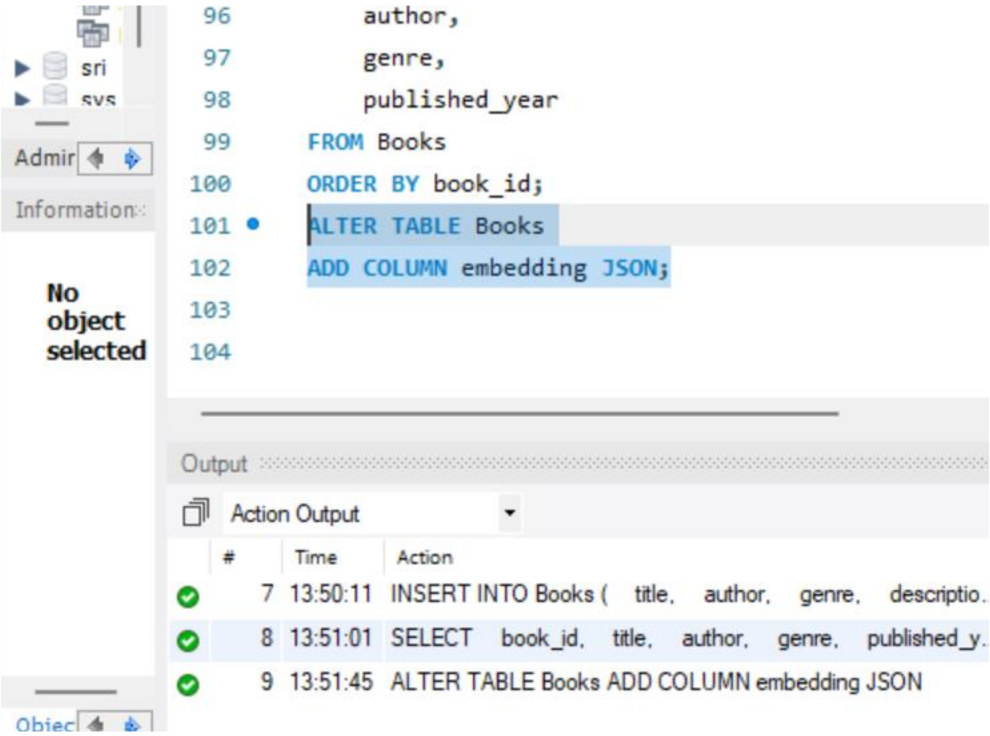



```
SELECT
  book_id,
  title,
  author,
  genre,
  published_year
FROM Books
ORDER BY book_id;
```

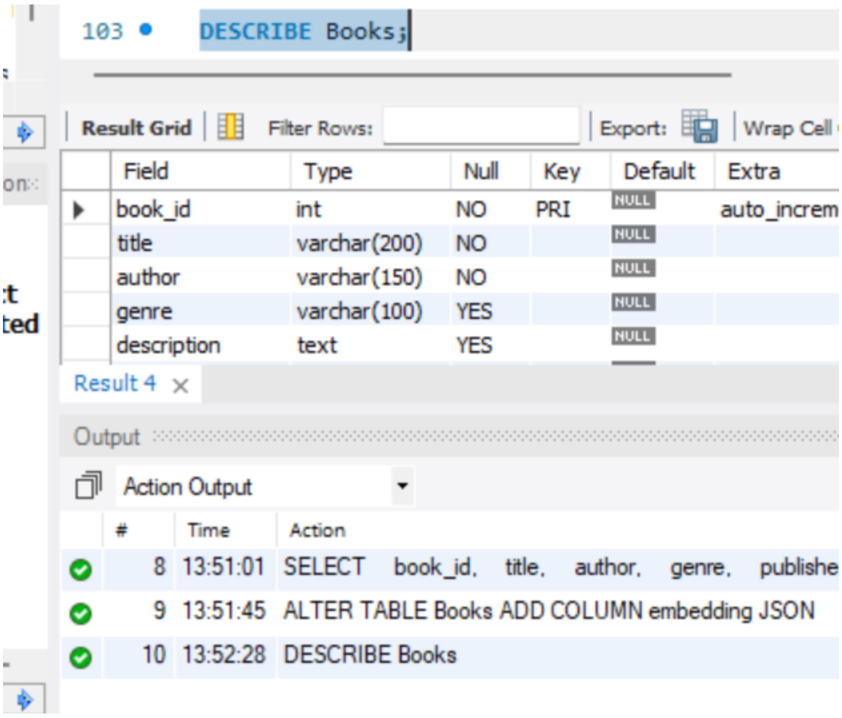


ALTER TABLE Books

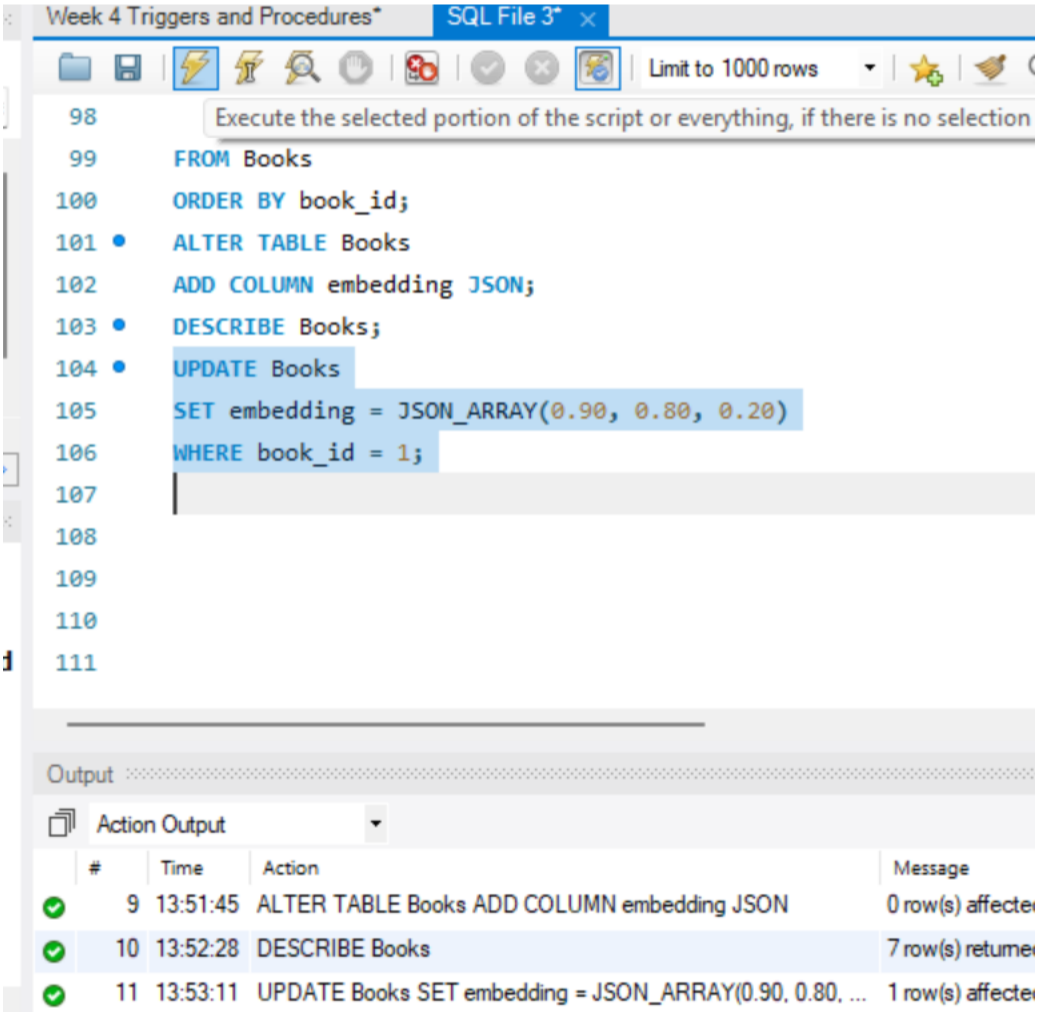
ADD COLUMN embedding JSON;



DESCRIBE Books;



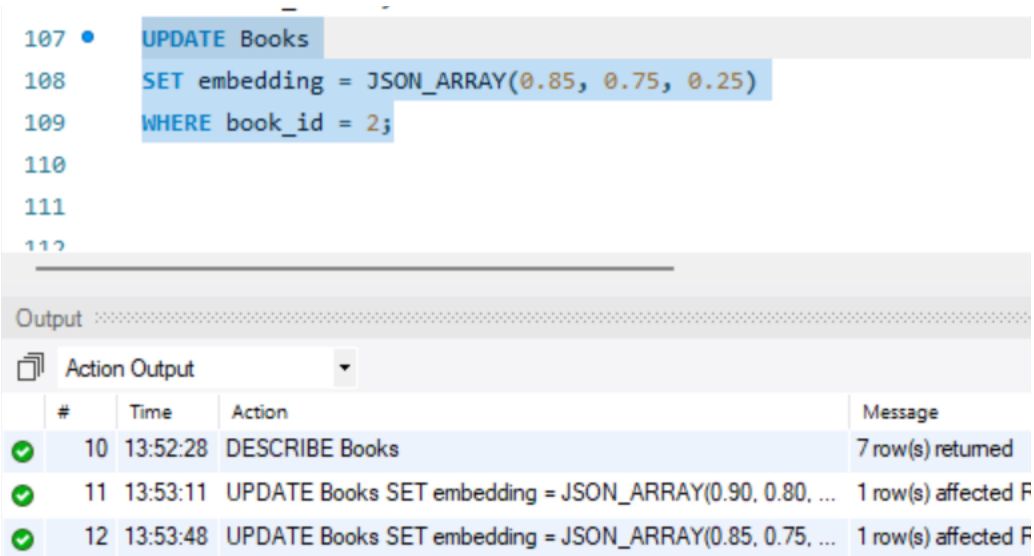

```
UPDATE Books

SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)

WHERE book_id = 1;
```



UPDATE Books

SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)

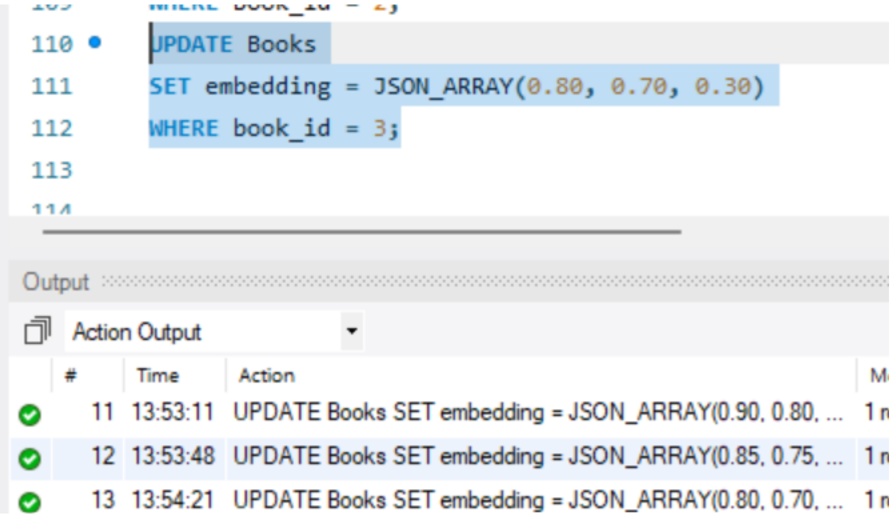
WHERE book_id = 2;



UPDATE Books

SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)

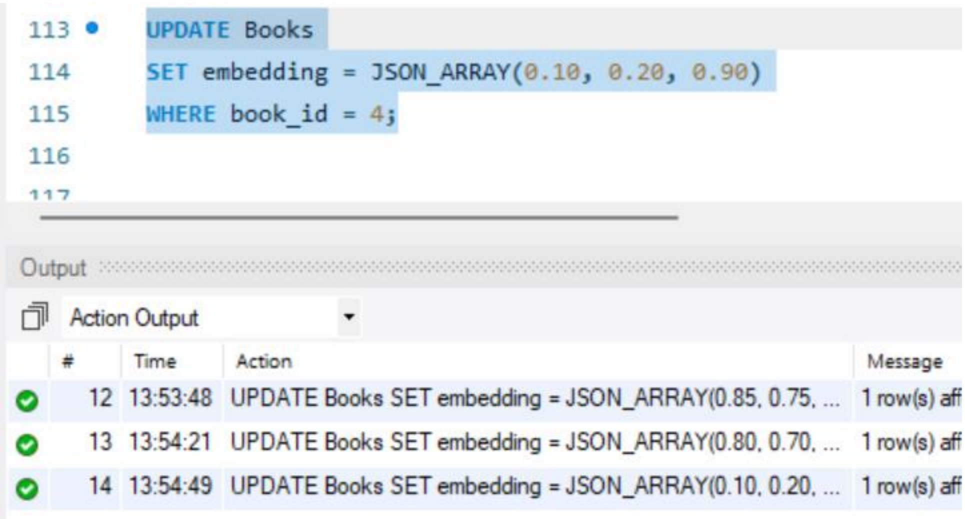
WHERE book_id = 3;



UPDATE Books

SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)

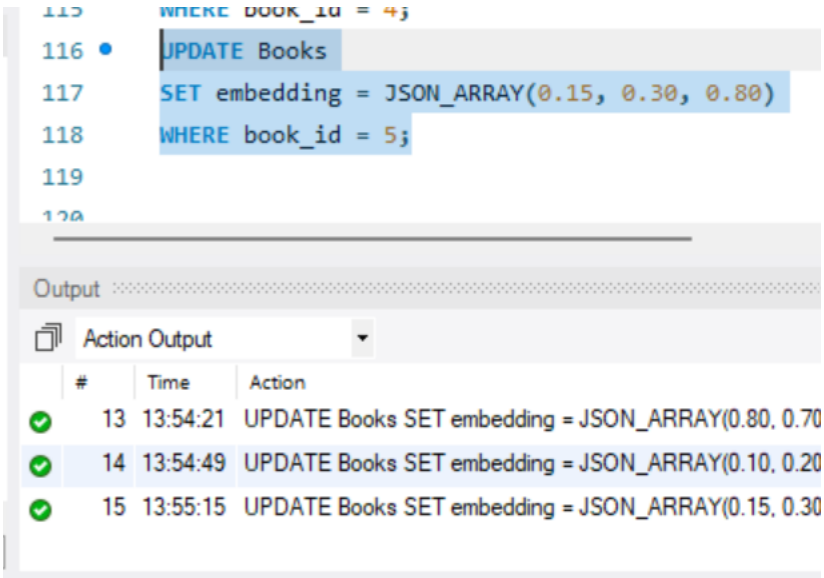
WHERE book_id = 4;



UPDATE Books

SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)

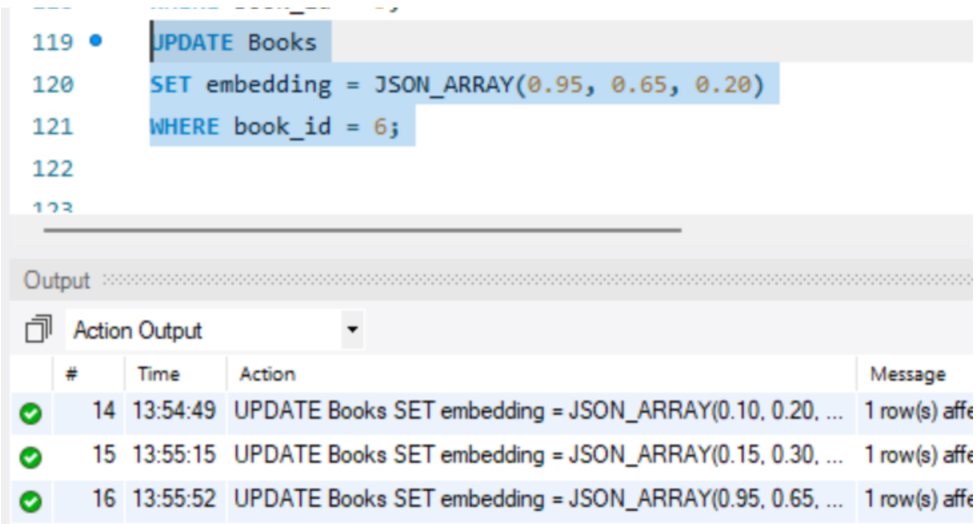
WHERE book_id = 5;



UPDATE Books

SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)

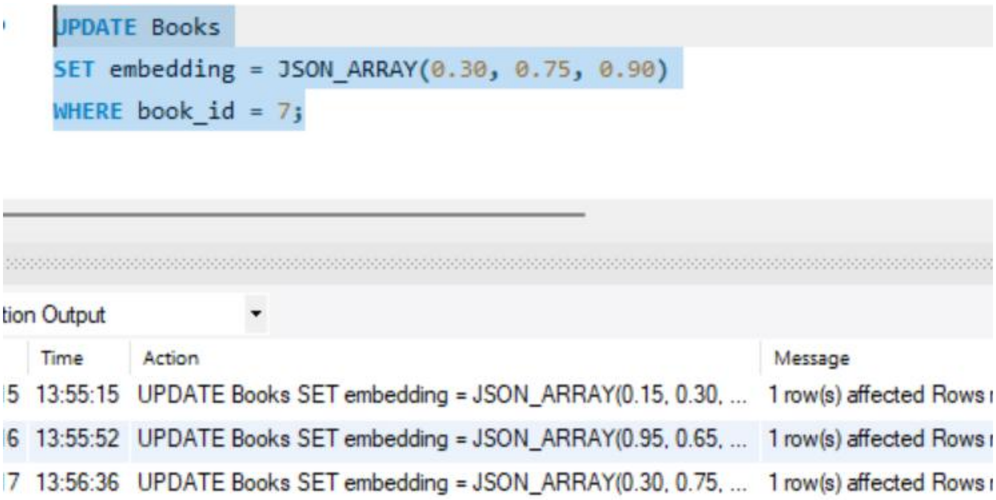
WHERE book_id = 6;



UPDATE Books

SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)

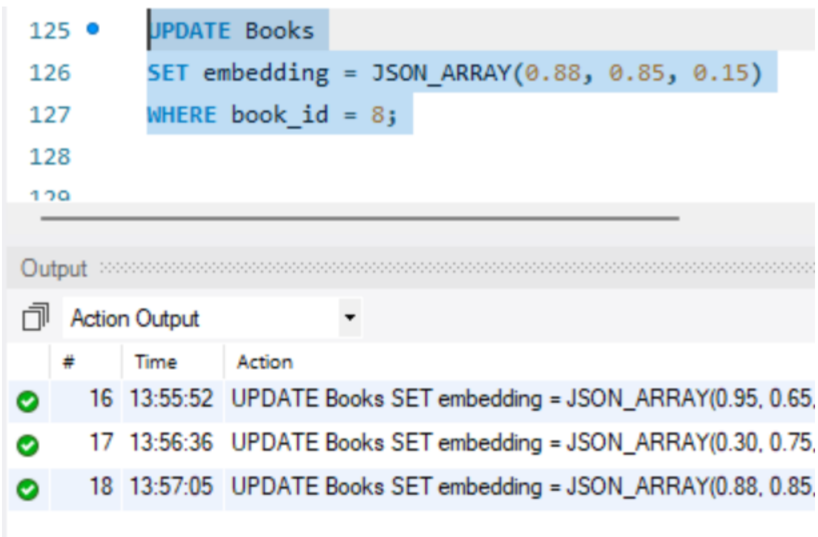
WHERE book_id = 7;



UPDATE Books

SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)

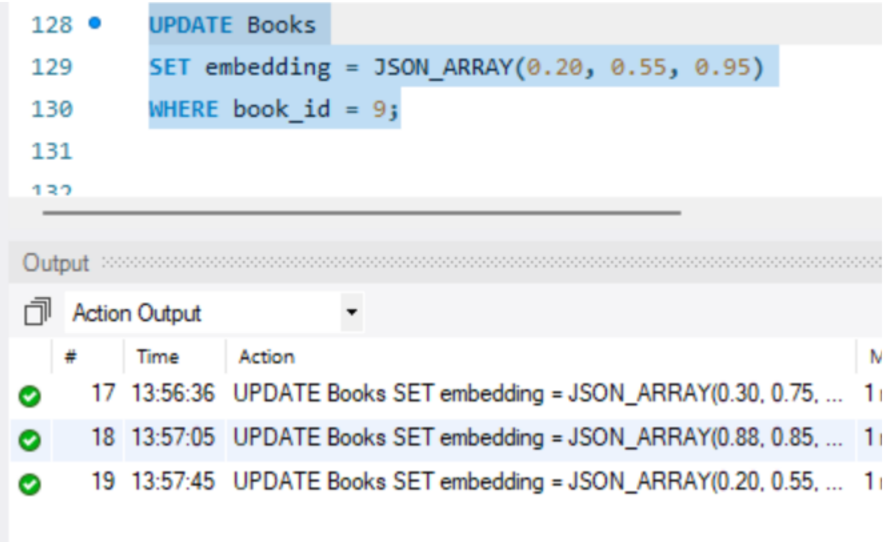
WHERE book_id = 8;



UPDATE Books

SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)

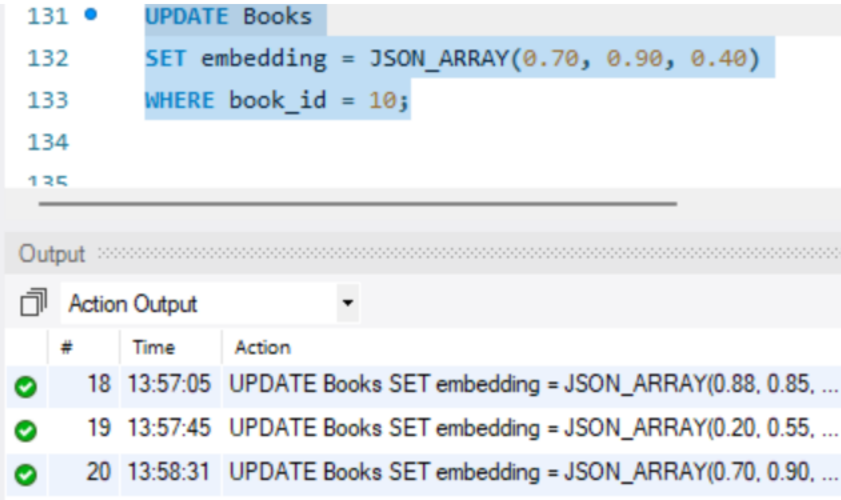
WHERE book_id = 9;



UPDATE Books

SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)

WHERE book_id = 10;



SELECT

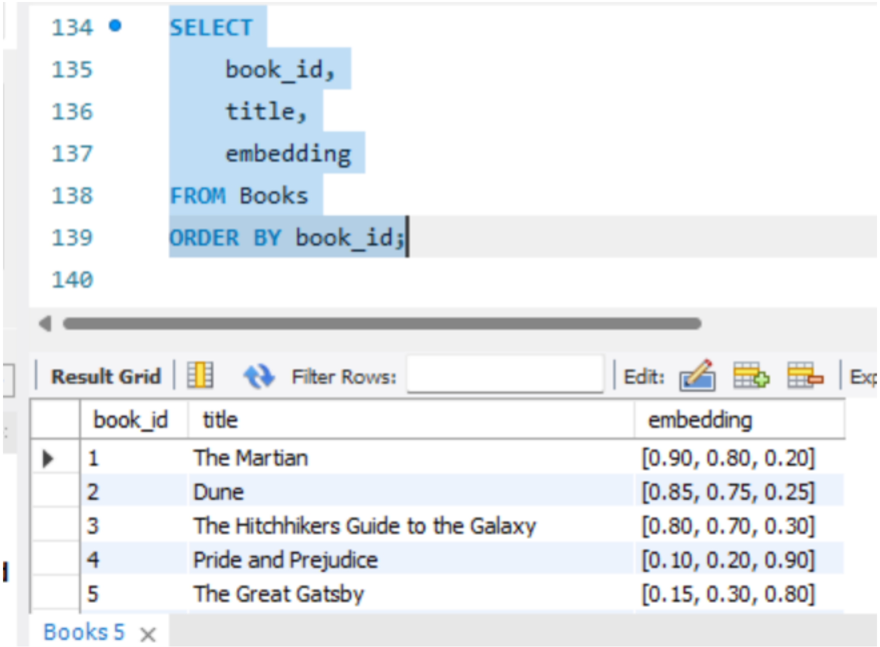
  book_id,

  title,

  embedding

FROM Books

ORDER BY book_id;



```sql
SELECT
    book_id,
    title,
    embedding
FROM Books
ORDER BY book_id;
```

Result Grid | Filter Rows: | Edit: | Exp

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]

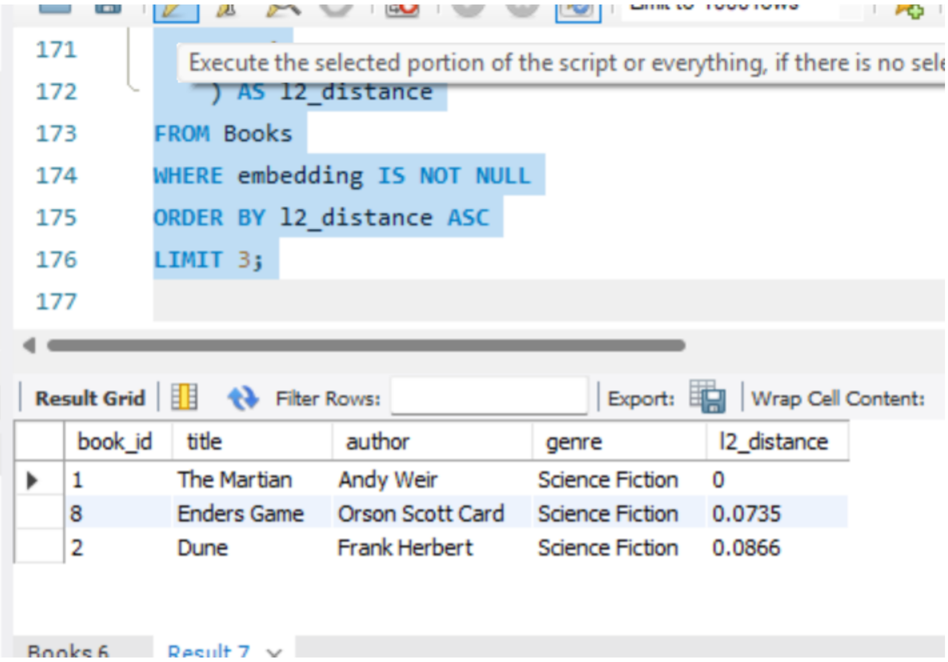
Books 5 ×

```sql
SELECT

  book_id,

  title,

  author,

  genre,

  ROUND(

    SQRT(

      POW(
        CAST(
          JSON_EXTRACT(embedding, '$[0]')
          AS DECIMAL(10,6)
        ) - 0.90,
        2
```

```
        )

        +

        POW(
          CAST(
            JSON_EXTRACT(embedding, '$[1]')
            AS DECIMAL(10,6)
          ) - 0.80,
          2
        )

        +

        POW(
          CAST(
            JSON_EXTRACT(embedding, '$[2]')
            AS DECIMAL(10,6)
          ) - 0.20,
          2
        )

      ),

      4

    ) AS l2_distance

FROM Books

WHERE embedding IS NOT NULL
```

ORDER BY l2_distance ASC

LIMIT 3;



SELECT

book_id,

title,

author,

genre,

ROUND(

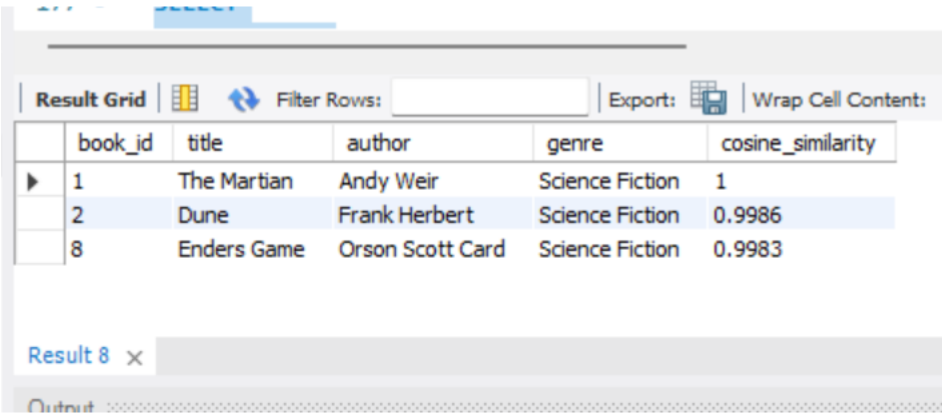
(

CAST(

```
    JSON_EXTRACT(embedding, '$[0]')
    AS DECIMAL(10,6)
  ) * 0.90

  +

  CAST(
    JSON_EXTRACT(embedding, '$[1]')
    AS DECIMAL(10,6)
  ) * 0.80

  +

  CAST(
    JSON_EXTRACT(embedding, '$[2]')
    AS DECIMAL(10,6)
  ) * 0.20

)

/

(

  SQRT(

    POW(
      CAST(
        JSON_EXTRACT(embedding, '$[0]')
        AS DECIMAL(10,6)
      ),
```

```
    2
  )

  +

  POW(
    CAST(
      JSON_EXTRACT(embedding, '$[1]')
      AS DECIMAL(10,6)
    ),
    2
  )

  +

  POW(
    CAST(
      JSON_EXTRACT(embedding, '$[2]')
      AS DECIMAL(10,6)
    ),
    2
  )

)

*

SQRT(

  POW(0.90, 2)
```

```
            +

            POW(0.80, 2)

            +

            POW(0.20, 2)

          )

        ),

        4

    ) AS cosine_similarity

FROM Books

WHERE embedding IS NOT NULL

ORDER BY cosine_similarity DESC
LIMIT 3;
```



book_id	title	author	genre	cosine_similarity
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986
8	Enders Game	Orson Scott Card	Science Fiction	0.9983